

Homework Submission

Student Name
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Problem Statement

Solve the quadratic equation

$$x^2 - 2x = 0.$$

Key Principles and Formulas

- A quadratic equation of the form $ax^2 + bx + c = 0$ can be solved by factoring, completing the square, or using the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- If the quadratic can be written as a product of linear factors, $(x - p)(x - q) = 0$, then each factor gives a root: $x = p$ or $x = q$.
- Completing the square rewrites $ax^2 + bx$ as $a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a}$, which is useful when factoring is not obvious.

Detailed Step-by-Step Solution

We solve the equation by two complementary methods.

Method 1: Factoring

$$\begin{aligned}x^2 - 2x &= 0 \\x(x - 2) &= 0 \\x = 0 \quad \text{or} \quad x - 2 &= 0 \\x = 0 \quad \text{or} \quad x &= 2\end{aligned}$$

Method 2: Completing the Square

$$\begin{aligned}x^2 - 2x &= 0 \\x^2 - 2x + 1 &= 1 && \text{(add 1 to both sides)} \\(x - 1)^2 &= 1 \\x - 1 &= \pm 1 \\x &= 1 \pm 1 \\x &= 2 \quad \text{or} \quad x = 0\end{aligned}$$

Both methods yield the same set of solutions.

Final Answer

$$\boxed{x = 0 \text{ or } x = 2}$$

Study Notes and Key Takeaways

- Always check whether a quadratic can be factored quickly; factoring avoids unnecessary algebra.
- When completing the square, remember to add the same constant to both sides to keep the equation balanced.
- After finding factors or a completed-square form, set each factor (or each side of the equation) equal to zero to obtain all possible roots.